

Katalon Script Generator — User Guide

This guide explains how to use the web application for every input tab and the shared options below them.

1. What the application does

The tool converts human-readable **test steps** (and optional **locators**) into **Katalon Studio Groovy** scripts for WebUI or Mobile keywords. Generation uses **Gosi Brain** on the server.

2. Running locally vs production

Local development

- Start **backend**: `npm run dev` in `server/` (default port **8787**).
- Start **frontend**: `npm run dev` in `client/` (default port **5173**).
- Open `http://localhost:5173`. API calls use the Vite proxy to the backend.

Production (example layout)

- Frontend** may be hosted on Netlify (static site).
- Backend** should run on a full Node host such as **Render** where **Playwright** can launch Chromium (locator extraction and browser recording).

Configure the frontend build with `VITE_API_URL` pointing at your backend URL (no trailing slash). On the backend, set `ALLOWED_ORIGINS` to your Netlify URL(s) for CORS.

3. Header status

- Gosi Brain: ready** — backend is reachable and chat URL + API key are configured.
- API unreachable** — frontend cannot reach `/api/health` (wrong `VITE_API_URL`, backend down, or network).
- Gosi Brain: set ... env vars** — backend runs but server-side credentials are missing.

Optional bearer token

If your organisation passes a JWT via WebView, open the app once with `?token=your_token` (or full `Bearer ...`). It is saved in the browser and sent with generate requests. If empty, the server may use `GOSI_BRAIN_AUTHORIZATION_TOKEN` from environment variables instead.

4. Tabs overview

| Tab | Purpose |

|----|-----|

| **Manual** | Type or paste steps yourself. |

| **CSV** | Import steps from a CSV file (simple rows or multi test-case export). |

| **Jira** | Fetch steps from a Jira issue description (optional credentials). |

| **Record** | Capture a Web flow in a real Chromium window on the **backend machine**; produces a Playwright-style script and locator lines. |

Note: **Record** is disabled when **Platform** is **Mobile** — recording uses Web-only Playwright.

5. Shared fields (below every tab)

These apply whichever tab is selected.

Platform

- **Web (WebUI)** — Standard browser automation keywords.
- **Mobile (Mobile)** — Requires **Appium** running locally (see section 10).

Gosi Brain model

Choose the model exposed by your server configuration (default is shown in the dropdown).

Test case name (optional)

Used when naming or exporting the generated test case. Leave blank if you do not need a fixed name.

Katalon project path (for export)

Absolute path to your **Katalon Studio project folder** on **your PC** (must contain a **Test Cases** directory). Used by **Add to Katalon Project** — this writes files **locally**, not on the server.

Format / style pass

- **None** — Minimal post-processing.
- **Simplify layout** — Cleans up formatting after generation.
- **Match project style** — Adjusts wording/layout toward your project context.

Locators

One line per mapping, for example:

```
Login button = #submit-login
Username field = input[name='user']
```

Names can align with steps so the compiler resolves targets. Manual lines **take priority** when merged with auto-detected lines.

Auto convert before generate

When enabled, Playwright/Selenium/Cypress-style locator lines are converted to **Katalon CSS/XPath** before generation.

Use **Convert to Katalon Locators** to preview the conversion table without generating yet.

Page URL (preview & auto-detect for generate)

HTTPS URL of the page under test. Used by:

- **Preview locators from URL** — fills the read-only Playwright-style preview panel.
- **Auto-detect for generate** — server visits the URL and merges detected locator candidates into generation (manual locators still win on conflicts).

Page language (Playwright extraction)

Auto, **English**, or **Arabic** — influences browser locale when extracting locators from the Page URL.

Auto-detect for generate

When checked (and Page URL is filled), the backend pulls locator candidates from the live page during **Generate**. Requires a backend with working **Playwright + Chromium** (set `PLAYWRIGHT_BROWSERS_PATH` on hosts like Render so browsers persist after build).

Stream response (live typing)

Streams tokens into the Groovy panel during generation instead of waiting for one response.

6. Manual tab

1. Enter **test steps**, **one per line**.
2. Optionally fill **Locators**, **Page URL**, enable **Auto-detect**, etc.
3. Click **Generate Katalon Groovy**.

Tip: Clear step-specific wording ("Click Login", "Type password") helps the compiler.

7. CSV tab

Upload

Choose a `.csv` file. The parser detects two shapes:

Simple format

Columns similar to `** Step **` and `** Description **` — each row becomes step text used for generation.

Test-case export format

Multiple test cases with a `** Steps **` column (for example Zephyr/VIC-style exports). Expected-result suffixes such as `** | Expected: ... **` are stripped.

After load:

- Use `**Select all**` / `**Clear all**` and row checkboxes to choose which test cases contribute steps.
- The status line shows how many steps will be sent to `**Generate**`.

Generate

Stay on CSV after selection and click `**Generate Katalon Groovy**` — steps come from selected rows.

8. Jira tab

Credentials (optional)

- `**Jira base URL**` — Site root only, e.g. `https://your-domain.atlassian.net` or `https://jira.company.com`.
- `**Email or username**` — Atlassian Cloud uses email; Server/Data Center may use username.
- `**Password, PAT, or API token**` — Same combination you would use for Basic auth or PAT-only REST where configured.

Leave all three empty to load `**demo**` steps only (nothing is stored on the server).

Buttons

- `**Test Jira login**` — Calls `** GET /myself **` to verify URL and credentials without loading an issue.
- `**Fetch from Jira**` — Loads the issue and parses steps into the `**Test steps from Jira**` box.

Editing steps

Edit the textarea if needed — `**Generate**` uses exactly these lines while you remain on the `**Jira**` tab.

Troubleshooting

- `**401**` — Wrong URL, user, or secret.
 - `**403**` — Often blocked Basic auth or missing REST permission; try a `**PAT**` or ask your Jira admin.
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9. Record tab (Web only)

Recording opens **Chromium on the machine running the backend**, not inside your laptop browser alone.

1. Enter **URL to open**.
2. Click **Record test flow** — interact with the browser window; finish via **Finish recording** on the page or wait for timeout.
3. The **Playwright-style script** textarea fills; locator **name = selector** lines merge into **Locators**.
4. Adjust locators if needed, then **Generate Katalon Groovy**.

Lossless replay

Check **Lossless replay** to minimise rewriting (preserve trace fidelity; skips some Groovy optimisers).

Stuck session

Use **Cancel recording on server** if you closed the browser or refreshed the UI while a session was active.

Paste-only workflow

You may paste an existing Playwright script instead of recording — still fill **URL** when the server needs it for locator extraction.

10. Mobile (Appium) panel

Shown when **Platform** is **Mobile**.

1. Start **Appium** locally (default `http://127.0.0.1:4723`).
2. Paste **Capabilities** as JSON matching your device/emulator.
3. **Check Appium** — Verifies connectivity.
4. **Start session** — Opens an Appium session.
5. **Extract locators** — Reads page source and fills locator suggestions into **Locators**.
6. **Stop session** — Ends the session.

Mobile recording proxy

Start Recording / **Stop Recording** uses a proxy URL shown in the hint — use it **only** in Appium Inspector or another client as directed; **do not** replace the main Appium URL field used for extract/stop.

Apply to Steps+Locators moves captured commands into Manual-oriented steps when recording stopped.

11. Output panel

- **Clear** — Clears generated Groovy text.
- **Copy** — Copies script to clipboard.
- **Add to Katalon Project** — Writes under **Test Cases** using **Katalon project path** (local disk).
- **Download .groovy** — Downloads the script file.

Lint warnings may appear below when the server validates Groovy.

12. History

Past generations may appear in **History** — click an entry to reload steps and script into the workspace.

13. Quick checklist before Generate

- Backend reachable (**Gosi Brain: ready**).
 - Valid bearer token if your deployment requires it (or server env fallback configured).
 - Steps present **or** Record tab has script/URL as required.
 - Locators filled or **Auto-detect** + **Page URL** if you rely on automatic mapping.
 - **Playwright features** only work where Chromium is installed on the backend (`npx playwright install chromium` during deploy and persistent browser path if applicable).
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14. Support commands (maintainers)

From repository root:

```
npm run docs:user-pdf
```

Rebuilds `docs/USER_GUIDE.pdf` from this Markdown file (requires Playwright/Chromium locally).